Project Overview

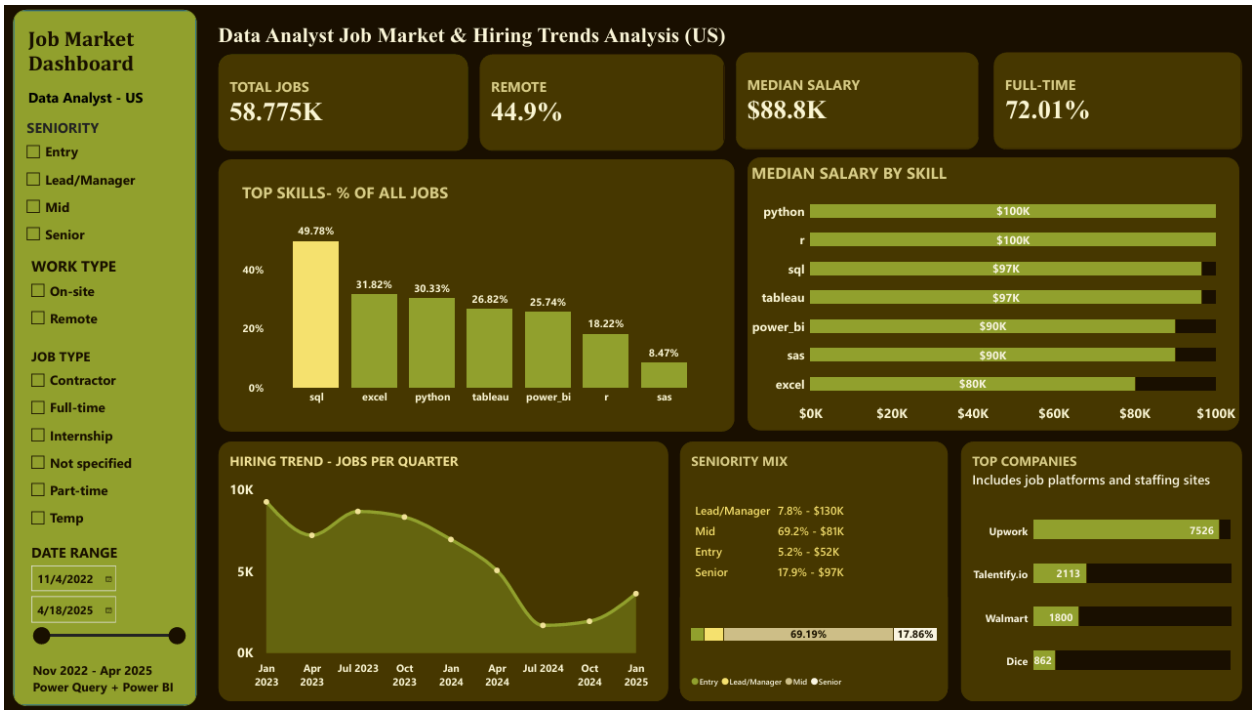
Title: Data Analyst Job Market & Hiring Trends Analysis (US)

Problem: I wanted to find out which skills, salaries and work types employers ask for in data analyst jobs, so I know what to focus on while learning.

Dataset: Kaggle, “Data Analyst Job Postings [Pay, Skills, Benefits]” by Luke Barousse. 58,775 US job postings, Nov 2022 to Apr 2025, after cleaning.

Tools: Power Query (data cleaning and transformation), Power BI (dashboard), DAX (measures).

Goal: Build an end-to-end analysis, from raw CSV to an interactive one-page dashboard.



Raw Data: Source file: Kaggle dataset “Data Analyst Job Postings [Pay, Skills, Benefits]” by Luke Barousse, CSV format, loaded from the Downloads\archive folder. Downloaded on 21 October 2026.

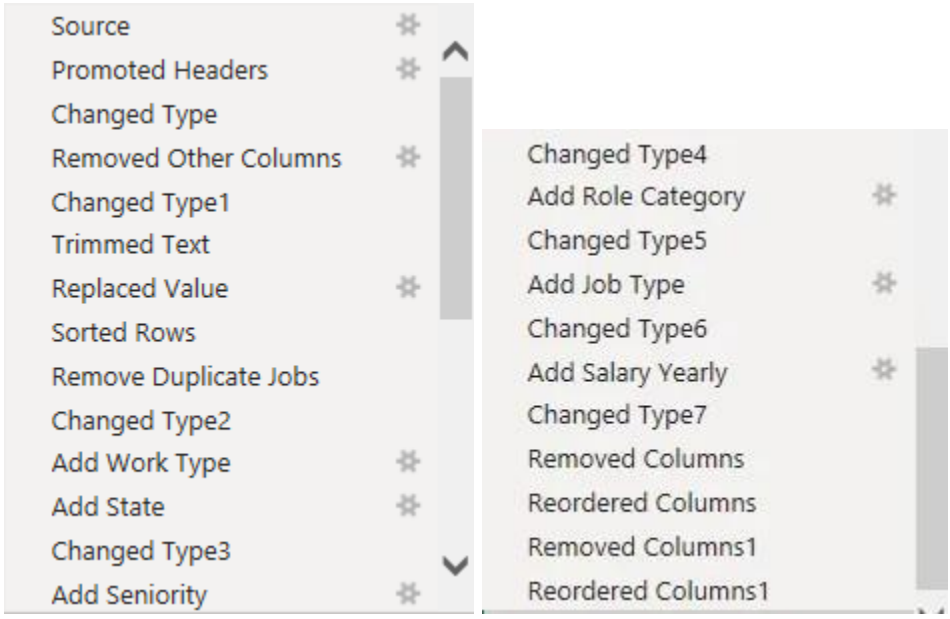
Before cleaning (raw):

- Rows: 61,954 job postings
- Columns: 27
- Problems I found:

- Duplicate job postings
- Blank or null salary values for many rows
- Skills stored in one cell as a list, e.g. `['python', 'sql', 'r']`
- Location in mixed formats, e.g. “Denver, CO”, “Anywhere”, “United States”, and extra text like “(+2 others)”
- Schedule type with many different values (33 distinct values in the cleaned data)
- Job titles written in many ways, e.g. “Sr. Data Analyst”, “Senior Data Analyst”, “Data analyst”
- Extreme salary values (outliers)

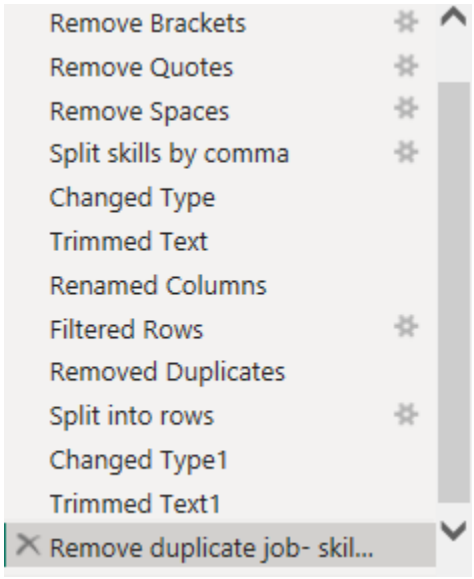
After cleaning: 58,775 jobs (Jobs query), and 189,280 job-skill rows (Skills query).

Jobs Query:
Applied Steps : (25 steps)



Skills:

Applied Steps :(15 steps)



Custom Columns

Column	Source column	Logic (simple English)	Example
Work Type	work_from_home	true = Remote, else On-site	true → Remote
State	location	Last part after the comma	“Denver, CO” → CO
Seniority	title	Keywords: Senior/Sr → Senior, Lead/Manager → Lead/Manager, Junior/Entry/Intern → Entry, others Mid	“Sr Data Analyst” → Senior
Role Category	title	Grouped into 7 groups by keywords	“Data Scientist” → Data Scientist
Job Type	schedule_type	Grouped into 6 types	“Full-time” → Full-time
Salary Yearly	salary_standardized	Below $20,000 or above $300,000 → null	Outliers set to null

Key Insights

- SQL appears in about 50% of jobs (49.78%). Excel (31.8%) and Python (30.3%) come next.
- Among the top 7 skills, Python and R have the highest median salary (about $100K). Excel has the lowest (about $80K).
- 44.9% of jobs are remote, and 72% are full-time.
- Median salary across all jobs is $88.8K.
- Salary rises with seniority: Entry $52K, Mid $81K, Senior $97K, Lead/Manager $130K.
- Entry-level jobs are only 5.2% of postings. Projects and practical skills matter for freshers.

Limitations

- US data only, salaries in US dollars. This is not an India job market analysis.
- Seniority is based on job titles. Titles without a clear keyword are set to Mid.
- Median salary uses only jobs that show a salary.
- Skill % is calculated out of all jobs, including jobs with no skills listed.
- The hiring trend chart does not show incomplete quarters. I cannot confirm whether the 2024 drop is a real decline or a data collection effect